CSE4203 Principles of Software Architecture

RapidMed

Emergency Triage and Real-Time Ambulance Dispatch System

Milestone 1: Proposal and architectural requirements

Task 7: Architecturally significant requirements (ASRs)

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Team	RapidMed
Course	CSE4203 Principles of Software Architecture
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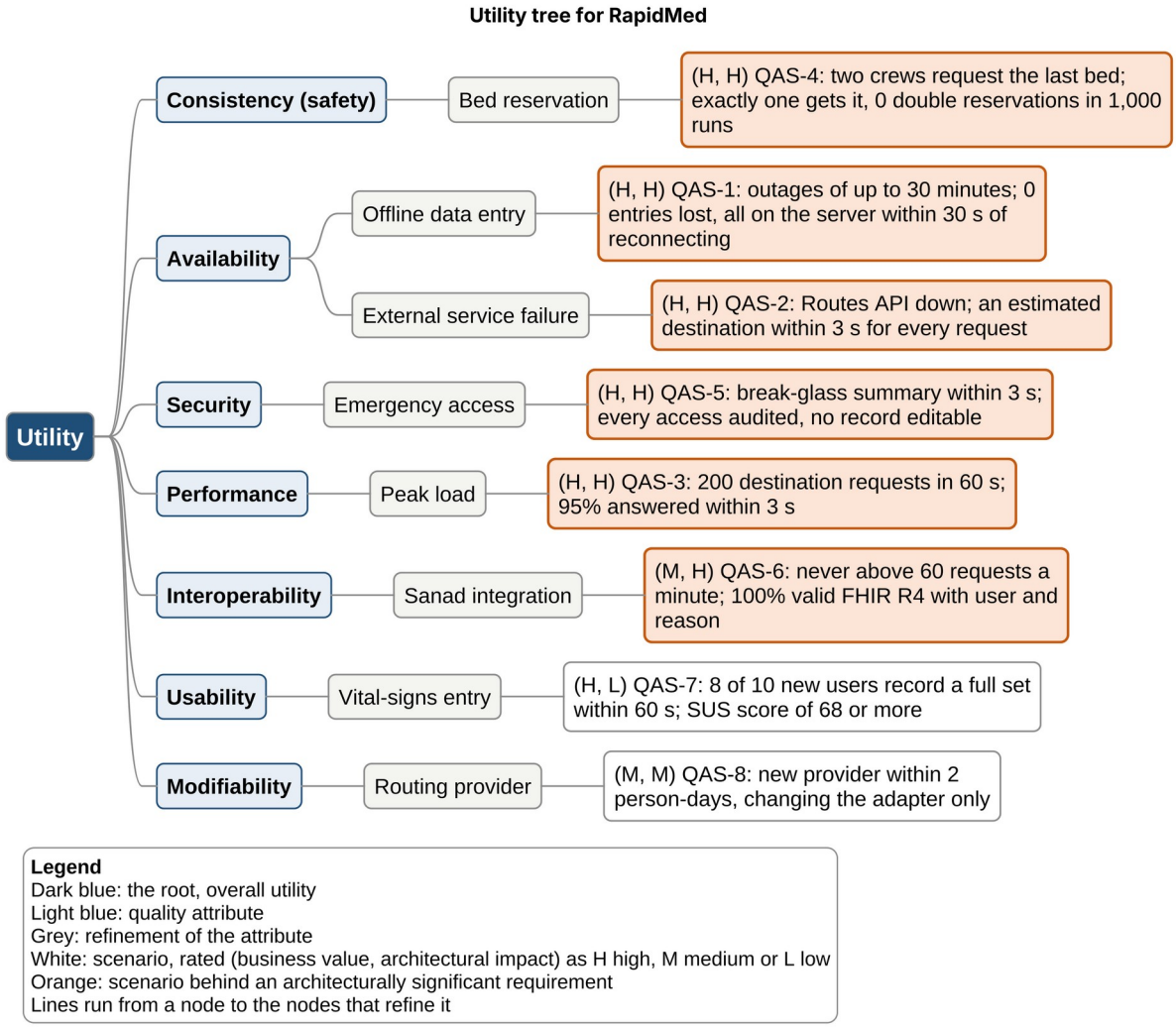
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7. Architecturally significant requirements (ASRs)

The utility tree places each quality attribute scenario under the quality attribute it refines, rated for business value and architectural impact. The table then ranks all the requirements, from the user stories (US), quality attribute scenarios (QAS) and constraints (C), on the same two scales: high (H), medium (M) or low (L). The five rated high for impact are the architecturally significant requirements, ASR-1 to ASR-5.



Rank	Requirement	From	Business value	Architectural impact
ASR-1	Bed reservations are all or nothing, even when requests arrive together or are retried	US-05, QAS-4	H	H
ASR-2	No data lost offline; queued entries sync in order, without duplicates	US-02, QAS-1	H	H
ASR-3	Break-glass access within 3 s, with an append-only audit log that can be reviewed	US-04, US-13, QAS-5, C7	H	H
ASR-4	A destination within 3 s at peak load, and an estimated one when the routing service fails	US-03, QAS-2, QAS-3, C5	H	H
ASR-5	Sanad reached only through FHIR R4, within its rate limit, behind a replaceable adapter	QAS-6, C2, C9	M	H
6	No patient data sent to Google services	C8, US-07	H	M
7	Overrides delivered to the crew and acknowledged	US-11, C3	H	M
8	Free service tiers only	C5, C6	M	M
9	Live map refreshed at least every 5 s	US-10	M	M
10	Routing provider replaceable within 2 person-days	QAS-8	M	M
11	Vital signs recorded in 6 taps or fewer each	US-02, QAS-7	H	L
12	The reasons behind each recommendation shown	US-14	L	L

Why these five will most influence the architecture

1. **ASR-1, atomic bed reservation.** It needs one authoritative record of reservations, updated with conditional writes, and requests that can be retried without effect. It also rules out reserving a bed offline, which puts consistency ahead of availability for that one action: the paramedic app has to separate what it may do offline (record, read cached data) from what must wait for the server (reserve).
2. **ASR-2, offline-first entry.** It needs a persistent store on the ambulance computer, an outbox that replays entries in order, and a server that recognises entries it has already received. Because several crews share the computer, the local store must also be encrypted and separated by user (US-01).
3. **ASR-3, break-glass with audit.** It needs an access path that skips consent but never skips the log, and an audit store kept apart from operational data, to which no application role can make changes. The same log supports the 72-hour breach duty (C7).
4. **ASR-4, a destination under load and failure.** It needs time-outs, a circuit breaker and a cache of travel times around the routing adapter. The free allowance of the Routes API (C5) makes the cache necessary even when the service is up.
5. **ASR-5, integration with Sanad.** It needs an adapter that keeps FHIR at the edge of the engine, with a client-side rate limiter and an episode cache, so that the mock and the real Sanad can be swapped. Its business value is medium because the prototype runs on the mock; its impact is high because it fixes where patient data enter RapidMed.